

Real-Time Basketball Jumpshot Detection on iOS: A Comparative Analysis of Linear Regression and Random Forest Classification Using Apple's Vision Framework

By: Davis Meng

Author Biography

Davis Meng is currently a sophomore at The Groton School in Massachusetts with demonstrated expertise in machine learning, computer vision, and iOS development. His research focuses on applying artificial intelligence to sports analytics, with published work in the Curieux academic journal on pose estimation for basketball shot analysis.

Abstract

Modern basketball increasingly demands the skill of precise shooting. However, accessible tools for analyzing shooting mechanics remain limited due to the lack of resources and reliability. This study addresses this gap by developing and comparing two machine learning approaches for real-time basketball jumpshot detection on consumer iOS devices. Building upon previous MediaPipe-based research that achieved basic success, this work transitions to Apple's native Vision framework. It leverages hardware-optimized pose detection to enable practical on-device analysis. Through systematic collection and annotation of basketball shooting footage, biomechanical features were extracted from body pose landmarks. These features capture the essential mechanics of any given jumpshot. Then, two contrasting machine learning architectures were developed and evaluated. Linear Regression was chosen for computational efficiency versus Random Forest for classification accuracy. They were evaluated for both predictive performance and real-world computational feasibility on mobile hardware.

The findings reveal a fundamental trade-off in mobile sports analytics. Linear Regression enables fluid, real-time feedback suitable for live training scenarios. Meanwhile, Random Forest provides substantially superior accuracy for post-session video analysis. Feature importance analysis consistently identified elbow positioning as the dominant biomechanical indicator across both models. This validates fundamental basketball coaching principles. Most significantly, thoughtful feature engineering based on domain knowledge produced dramatic improvements over generic statistical approaches. It demonstrates that understanding sport-specific mechanics outweighs raw computational power.

This research establishes that sophisticated basketball shot analysis can operate entirely on smartphones without specialized equipment. This could potentially democratize access to personalized coaching feedback for athletes at all levels.

Keywords: iOS Development, Apple Vision Framework, Sports Analytics, Pose Detection, Linear Regression, Random Forest, Real-Time Classification, Basketball Shot Detection, Mobile Machine Learning, On-Device Inference

Introduction

The evolution of mobile computing has fundamentally transformed what is possible in sports training. Smartphones now possess computational capabilities that rival those of desktop computers from just a few years ago. This allows for sophisticated machine learning applications to run entirely locally on the device. For sports analytics, this shift from cloud-based to local processing offers transformative advantages. These include real-time feedback with minimal latency, enhanced privacy through local data processing, elimination of internet connectivity requirements, and dramatically lower operational costs, having no need to run cloud services.

This study explores how these technological advances enable practical basketball shot detection and analysis on iOS devices. Building upon previous research utilizing MediaPipe for pose estimation (Meng 2024), this work leverages Apple's native Vision framework. This framework is specifically optimized for iOS hardware through the Neural Engine, as it is used to overcome the performance limitations inherent in cross-platform solutions.

The central challenge in mobile sports analytics lies in balancing classification accuracy against computational constraints. Basketball jumpshot detection must process video at 30 to 60 frames per second, requiring extracting pose landmarks, engineering biomechanical features, and performing model inference. All of this must happen while maintaining responsive real-time feedback. This study directly compares two fundamentally different approaches to this challenge. Linear Regression offers computational simplicity and speed. Random Forest Classification provides superior accuracy through ensemble learning at a higher computational cost.

This research investigates three core questions. How do Linear Regression and Random Forest models compare for jumpshot detection accuracy? Can either approach achieve true real-time performance on modern iOS devices? Which biomechanical features extracted from pose data most reliably predict jumpshot events? By answering these questions, this study establishes practical guidelines for deploying machine learning based sports analytics on mobile platforms, emphasizing the accuracy efficiency trade-offs that govern real-world deployment decisions.

Literature Review

The application of machine learning pose estimation to sports analytics has expanded rapidly in recent years, though significant challenges remain. Roggio et al. (2024) conducted a narrative review of major pose estimation models, including OpenPose, MediaPipe, BlazePose, and HRNet, across clinical, sports, and ergonomic contexts. While they found compelling potential

for non-invasive, cost-effective biomechanical assessment, the review identified persistent challenges in accuracy, data quality, and integration into existing practices, concluding that standardized evaluation frameworks remain essential for reliable applications. Complementing this broad survey, Dedhia et al. (2024) evaluated MediaPipe BlazePose as a virtual gym assistant and found it achieved accuracy within 10% of IMU-based motion capture for curated fitness movements, demonstrating practical applicability in training contexts. However, the authors noted limitations in complex motion scenarios, suggesting that more dynamic movements like those in competitive sports may demand further refinement.

The gap between controlled and competitive environments has been well documented. Cronin et al. (2023) tested OpenPose on footage from the 2017 World Athletics Championships long jump finals and found poor agreement with manual analysis, producing an average ICC (Intraclass Correlation Coefficient) of only 0.17 across variables, concluding that the framework is unsuitable for in-competition biomechanical analysis. In response to such accuracy limitations, Xi et al. (2024) developed a dual-channel spatiotemporal transformer architecture integrated with IoT devices for sports training, achieving mean per joint position scores of 42.2 mm and 29.1 mm on the Human3.6M and MPI-INF-3DHP datasets, respectively, significantly outperforming existing methods. Their work represents an important future direction for capturing temporal dynamics in movement analysis.

Despite limitations in competitive settings, validation studies have demonstrated that pose estimation can achieve acceptable accuracy for specific, well-defined tasks. Ino et al. (2024) compared OpenPose against 3D motion analysis and human manual analysis for measuring knee valgus during drop vertical jumps, finding no significant difference between AI-based and human-based methods (mean absolute errors of 2.4° and 3.2°, respectively). Fukushima et al. (2024)

provided broader cross-sport validation by comparing OpenPose against marker-based motion capture across seventeen athletic movements, including basketball chest passes. Their results showed systematic joint angle errors during rapid or complex movements, reinforcing the need for sport-specific validation rather than reliance on general accuracy benchmarks. Tharatipyakul et al. (2024), reviewing 45 articles on pose estimation feedback systems, found that CNNs (Convolutional Neural Networks) dominated pose detection while assessment methods ranged from mathematical formulas to machine learning classification, and observed a notable shift from OpenPose to MediaPipe as the most widely adopted framework.

The present study builds directly upon the author's prior research (Meng, 2024), which used MediaPipe's 33-landmark model and Random Forest classification to predict basketball shotmaking accuracy from single-camera video. That study achieved 100% training accuracy but only 51.1% test accuracy on 126 annotated clips, identifying key limitations: the insufficiency of static statistical feature aggregation for capturing shooting dynamics, limited dataset size, and processing times incompatible with real-time feedback. The current work addresses these gaps by transitioning to Apple's native Vision framework (Apple Developer Documentation, 2023), which leverages the Neural Engine for hardware-optimized pose detection of 17 body landmarks. Though fewer than MediaPipe's 33 points, this reduced landmark set, combined with domain-informed biomechanical feature engineering, over 8,000 annotated frames, and a comparative evaluation of Linear Regression versus Random Forest, yields both substantially improved accuracy and practical guidelines for the accuracy–efficiency trade-offs governing real-time mobile deployment. No prior study has systematically quantified these trade-offs for basketball shot detection on consumer iOS hardware, establishing the gap this research fills.

Methodology

Data Collection and Annotation

Video footage was captured using the iPhone's rear camera at 1080p resolution and 30 frames per second. The camera was positioned approximately 12 feet from the shooting location at chest height. Eight video sessions were recorded. Each contained 10 to 15 jumpshot attempts interspersed with non-shooting movements. These included walking, dribbling, and standing. This variety was essential to train models for distinguishing actual jumpshots from other basketball-related activities.

Manual temporal annotation established ground truth labels. Jumpshot events were defined from initial knee flexion through ball release. Annotations were compiled into CSV (Comma Separated Value) files with precise timestamp ranges. This ultimately yielded 8,410 annotated frames comprising 2,541 jumpshot frames and 5,869 non-jumpshot frames.

Pose Detection with Apple's Vision Framework

Unlike the MediaPipe implementation from previous research, this study utilized Apple's Vision framework. This provides 17 body landmarks. These are the nose, eyes, ears, shoulders, elbows, wrists, hips, knees, and ankles. The Vision framework processes frames asynchronously through a specific API and returns normalized coordinates for each detected landmark along with confidence scores.

A Python extraction pipeline processed annotated video segments frame by frame. It stored landmark coordinates with associated metadata, including clip ID, frame number, timestamp, and ground truth label.

Biomechanical Feature Engineering

Raw landmark coordinates alone provide insufficient discriminative power. This is because absolute positions vary with camera placement, shooter height, and distance. To address this, 19

engineered features were computed from raw landmarks. These were designed to capture biomechanical patterns invariant to these factors. Static spatial features included arm angles, normalized wrist and head heights, elbow displacement components, knee bend angles, body center position, head tilt, shoulder width, and body lean. Temporal dynamic features captured movement patterns. This was done through velocity and acceleration measurements of key body points across rolling time windows.

All spatial features were normalized using torso length as a reference measurement. This enabled generalization across different shooters and camera configurations. Missing landmark data occurred in approximately 3% of frames due to occlusion. This was handled through linear interpolation with forward and backward filling.

Model Training and Comparison

The complete dataset was split 70/30 into training and testing sets. Stratified sampling was used to maintain class distribution. Feature standardization was applied using StandardScaler. Parameters were fitted exclusively on training data to prevent data leakage.

Linear Regression was implemented using scikit learn's LinearRegression class without regularization. Despite the binary classification task, this approach was selected for its computational simplicity and interpretability. Predictions were thresholded at 0.5 for binary classification. Performance was evaluated using the R^2 score alongside traditional classification metrics.

Random Forest Classification utilized scikit learn's RandomForestClassifier with 100 estimators and unrestricted depth. This ensemble approach aggregates predictions from 100 decision trees trained on bootstrap samples. This provides robustness to overfitting while

maintaining interpretability through feature importance scores. Both models were evaluated on accuracy, precision, recall, F1 score, and 10-fold cross-validation performance.

iOS Integration and Performance Benchmarking

Models were exported for iOS integration. Linear Regression was exported as JSON containing coefficients and scaler parameters. Random Forest was exported as Core ML format. It was automatically converted to optimized Swift code by Xcode.

The iOS application implements a complete processing pipeline. This includes video capture at 30 FPS through AVFoundation, pose detection via Vision framework, feature extraction in Swift, model inference, and event clustering. Clustering groups of consecutive high probability frames. Performance was measured on iPhone 13 Pro using Xcode's Instruments profiling tool. It captured per-frame processing time, sustained frame rate, memory usage, battery consumption, model size, and loading time.

Results and Discussion

Model Performance Comparison

The Random Forest Classifier substantially outperformed Linear Regression across all metrics. Random Forest achieved strong test accuracy with high precision and good recall. Linear Regression showed modest performance with R^2 scores indicating limited explanatory power. The 10-fold cross-validation confirmed Random Forest's robust generalization with minimal variance.

Linear Regression's relatively low accuracy suggests that jumpshot classification involves substantial nonlinear relationships. These are not captured by linear models. The Random Forest model's perfect training accuracy initially raised overfitting concerns. However, strong test performance and cross-validation results indicate genuine learning of generalizable patterns.

Feature Importance Insights

Both models consistently identified elbow horizontal position as the dominant predictive feature. Its importance scores far exceeded those of other variables. This finding validates fundamental basketball coaching principles. These principles emphasize proper elbow alignment during shooting form. Body center vertical position, elbow vertical position, and head tilt provided secondary discriminative power.

Notably, spatial features capturing body configuration at individual time points dominated over temporal features. Temporal features measure movement patterns across frames. This suggests that jumpshot classification relies more on specific body positioning than on movement dynamics. This is an insight with important implications for feature engineering in similar applications.

Computational Performance Analysis

Linear Regression achieved genuine real-time performance on iPhone 13 Pro. It processed frames with minimal latency. The breakdown is 35ms for pose detection, 9ms for feature extraction, and just 3ms for inference. This enables 21 frames per second sustained processing. It is suitable for live training applications with responsive feedback.

Random Forest's processing time totaled 89ms per frame. This was dominated by 45ms inference time. This reflects the computational expense of evaluating 100 decision trees. This yields 11 FPS sustained performance. This falls short of smooth real-time requirements but is acceptable for offline video analysis.

Model sizes differed dramatically. Linear Regression was just 12 KB versus Random Forest at 847 KB. However, both remain negligible in modern application contexts. Battery

consumption showed that Linear Regression consumed half the power of Random Forest during continuous operation.

Comparison with the MediaPipe Approach

Previous research using MediaPipe achieved a limited test accuracy of 51.1%. It used 33 landmarks and statistical feature aggregation (Meng 2024). The current iOS implementation with Random Forest achieved 92.9% accuracy. This is an 81% improvement. It used fewer landmarks, 17 versus 33, but with domain-informed biomechanical features.

This dramatic improvement demonstrates a fundamental principle in applied machine learning. Thoughtful feature engineering based on domain knowledge yields far greater performance gains than incremental advances in model architecture or raw data quantity. The shift from generic statistical aggregates to biomechanical features capturing basketball shooting mechanics proved transformative.

Practical Trade-offs and Use Case Recommendations

The results establish clear use case recommendations. Linear Regression excels for real-time feedback applications. These require immediate shot detection to trigger coaching cues. It is also good for battery-constrained scenarios during extended practice sessions, and for resource-limited devices or embedded systems.

Random Forest is optimal for offline video analysis during post-practice review. It suits high-accuracy requirements where minimizing false positives is critical. It is also good for professional training contexts where accuracy justifies computational cost.

A hybrid approach combining both models shows promise. Linear Regression could perform initial real-time detection with a low threshold. This would be followed by Random Forest

verification at a higher threshold for confirmed events. This architecture could achieve approximately 15 to 18 FPS while maintaining Random Forest level accuracy for detected events.

Conclusion

Summary of Findings

This research successfully demonstrated the feasibility of real-time basketball jumpshot detection on iOS devices using Apple's Vision framework. Two contrasting machine learning approaches were developed, trained on over 8,000 annotated frames, and deployed on iOS hardware for comprehensive evaluation.

Random Forest Classification achieved superior accuracy with high precision and good recall. It substantially outperformed Linear Regression, which showed modest but interpretable results. However, Linear Regression achieved true real-time performance. Meanwhile, Random Forest's processing demands limited it to offline analysis applications.

Feature importance analysis revealed that elbow horizontal position dominates jumpshot prediction across both models. This validates coaching principles emphasizing proper shooting form alignment. Most significantly, domain-informed biomechanical feature engineering produced an 81% improvement over generic statistical approaches from previous work.

Theoretical Contributions

This work advances sports analytics research through several avenues. It provides platform-specific optimization, establishing iOS native performance benchmarks. It offers an empirical model selection framework quantifying accuracy efficiency trade-offs. It demonstrates that domain-informed feature engineering outperforms raw data quantity.

Practical Applications

The developed technology enables several immediate applications. These include real-time shooting form feedback during individual training, automated shot counting and tracking, post-practice video analysis with automatic shot detection, and remote coaching through analyzed video submissions.

Extended applications include integration with smart basketball equipment, team-level analytics aggregating multiple player data, injury prevention through biomechanical anomaly detection, and gamification of training through automated statistics.

Future Directions

Future research should investigate several areas. These include sequential modeling approaches like LSTMs (Long Short-term memory), Temporal Convolutional Networks, and Transformers to capture temporal dependencies. It should integrate ball tracking for release point and trajectory analysis. It should handle multi-person scenarios, dealing with occlusion and player tracking. It should explore shot outcome prediction based on biomechanical features. Research should work on generalization across diverse shooters and skill levels. Finally, it should pursue edge computing optimization through model quantization and Neural Engine utilization.

Closing Perspective

This research establishes that sophisticated basketball shot analysis can operate entirely on consumer smartphones without specialized equipment. The dramatic improvement over previous cross-platform approaches demonstrates that platform-specific optimization and domain-informed feature engineering substantially outweigh incremental model improvements.

The practical deployment of this technology on mobile devices represents a significant step toward democratizing sports analytics in training. While traditional motion capture systems cost tens of thousands of dollars and require dedicated facilities, smartphone-based analysis requires

270 only a device most athletes already own. This accessibility could transform skill development
271 processes and practice, particularly so in under-resourced communities where professional
272 coaching and advanced equipment remain inaccessible.

References

- Apple Developer. (2023). *Detecting human body poses in images*. Apple Developer Documentation. <https://developer.apple.com/documentation/vision/detecting-human-body-poses-in-images>
- Cronin, N. J., Walker, J., Tucker, C. B., Nicholson, G., Cooke, M., Merlino, S., & Bissas, A. (2024). Feasibility of OpenPose markerless motion analysis in a real athletics competition. *Frontiers in Sports and Active Living*, 5, Article 1298003. <https://doi.org/10.3389/fspor.2023.1298003>
- Dedhia, U., Bhoir, P., Ranka, P., & Kanani, P. (2023). Pose estimation and virtual gym assistant using MediaPipe and machine learning. In *2023 International Conference on Network, Multimedia and Information Technology (NMITCON)* (pp. 1–7). IEEE. <https://doi.org/10.1109/NMITCON58196.2023.10275938>
- Fukushima, T., Blauberger, P., Guedes Russomanno, T., & Lames, M. (2024). The potential of human pose estimation for motion capture in sports: A validation study. *Sports Engineering*, 27, Article 19. <https://doi.org/10.1007/s12283-024-00460-w>
- Ino, T., Samukawa, M., Ishida, T., Wada, N., Koshino, Y., Kasahara, S., & Tohyama, H. (2024). Validity and reliability of OpenPose-based motion analysis in measuring knee valgus during drop vertical jump test. *Journal of Sports Science and Medicine*, 23(1), 515–525. <https://doi.org/10.52082/jssm.2024.515>
- Meng, D. (2024). Predicting basketball shotmaking accuracy using single camera pose detection and random forest classification. *Curieux Academic Journal*. <https://www.curieuxacademicjournal.com>
- Roggio, F., Trovato, B., Sortino, M., & Musumeci, G. (2024). A comprehensive analysis of the machine learning pose estimation models used in human movement and posture analyses: A narrative review. *Heliyon*, 10(21), Article e39977. <https://doi.org/10.1016/j.heliyon.2024.e39977>
- Tharatipyakul, A., Srikaewsiew, T., & Pongnumkul, S. (2024). Deep learning-based human body pose estimation in providing feedback for physical movement: A review. *Heliyon*, 10(17), Article e36589. <https://doi.org/10.1016/j.heliyon.2024.e36589>

302 Xi, X., Zhang, C., Jia, W., & Jiang, R. (2024). Enhancing human pose estimation in sports
303 training: Integrating spatiotemporal transformer for improved accuracy and real-time
304 performance. *Alexandria Engineering Journal*, 109, 144–156.
305 <https://doi.org/10.1016/j.aej.2024.08.072>